

ENVIRONMENTAL SCIENCE

BHM4T02 · NEP 2020 · RTMNU / GNIET

B.Tech CSE (Data Science) · Semester IV

UNIT 1: AIR POLLUTION AND ITS CONTROL TECHNIQUES

■ Total Credits 02 T	■ University Exam 70 Marks	■ College Assessment 30 Marks	■ Lectures 6 Hours (Unit 1)
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■ SYLLABUS COVERAGE

- ✓ Contaminant behaviour in the environment
- ✓ Air pollution due to SO_x, NO_x, Photochemical Smog, Indoor Air Pollution
- ✓ Natural pathways for degradation: Carbon Cycle, Sulphur Cycle, Nitrogen Cycle, Oxygen Cycle
- ✓ Factors altering atmosphere: Deforestation, Fossil Fuels, Industrial & Vehicular Emissions, CFCs
- ✓ Techniques to control Air Pollution (Cyclone, ESP, Bag Filter, Wet Scrubber, Catalytic Converter)
- ✓ Ambient Air Quality & Continuous Air Quality Monitoring (NAAQS, AQI, CAAQM)
- ✓ Control measures at source, Kyoto Protocol & Carbon Credits

Includes RTMNU 7-Mark OR-Type Questions with Answer Guides

SECTION 1: CONTAMINANT BEHAVIOUR IN THE ENVIRONMENT

1.1 Definition of a Contaminant

A **contaminant** is any physical, chemical, biological, or radiological substance present in the environment — air, water, or soil — at concentrations sufficient to cause harm to human health, living organisms, or the overall ecosystem. Contaminants differ from pollutants in that a contaminant is simply a substance out of place, while a pollutant is a contaminant that actively causes measurable damage.

1.2 Sources of Contaminants

■ A. Industrial & Commercial Facilities

Factories, chemical plants, refineries, and mining operations release toxic wastes — heavy metals (lead, mercury, cadmium), acids, solvents — into air, water, and soil through exhaust stacks, effluent channels, and improper dumping. These are classified as **point sources** because pollution originates from a single identifiable location.

■ B. Oil & Chemical Spills

Accidental spills from oil tankers, pipelines, or offshore drilling platforms contaminate oceans, coastlines, and groundwater with petroleum hydrocarbons (benzene, toluene, xylene) and toxic chemicals. These are persistent organic pollutants (POPs) that resist natural degradation.

■ C. Non-Point Source Pollution (NPS)

Non-Point Source Pollution is contamination that **cannot be traced to a single discharge point**. Stormwater runoff from agricultural fields, roads, and parking lots carries fertilisers, pesticides, and oils into water bodies. NPS is the leading global cause of water quality degradation and is extremely difficult to regulate.

■ **NOTE:** NPS vs Point Source: Point source = single identifiable outlet (factory pipe, sewage drain). NPS = diffuse, spread over large areas (agricultural runoff, urban stormwater). NPS is harder to control because it has no single origin.

■ D. Sewage Systems & Wastewater Plants

Improperly treated municipal sewage discharges pathogens (bacteria, viruses), heavy metals, and high BOD organic matter into rivers and coastal waters. In India, approximately **80% of sewage is discharged untreated**.

■ E. Hazardous Waste Sites

Old industrial sites, landfills, and illegal dumping zones leach persistent contaminants — heavy metals, chlorinated solvents, radioactive materials — into groundwater and surrounding soil for decades.

1.3 Key Contaminant Behaviour Processes

■ Bioaccumulation

Bioaccumulation is the gradual build-up of a contaminant in an organism's body at a rate faster than it can be eliminated. Persistent organic pollutants (POPs) like DDT and heavy metals like mercury dissolve in fatty tissues and accumulate over an organism's lifetime. A fish in mercury-contaminated water accumulates far more mercury in its body than the surrounding water contains.

■ Biomagnification

Biomagnification is the progressive *increase* in concentration of a contaminant at each successive trophic level of the food chain. A small amount of DDT in phytoplankton becomes higher in zooplankton, higher still in small fish, and highest in apex predators (eagles, humans) — often **thousands of times higher** than the original environmental concentration. This is why predatory fish like tuna and swordfish carry dangerous mercury levels.

■ **NOTE:** DDT was banned in India in 2020. Mercury from coal power plants enters oceans → bioaccumulates in phytoplankton → biomagnifies through the food chain → reaches dangerous levels in tuna and swordfish consumed by humans.

■ **EXAM TIP:** 7-Mark Q: Explain contaminant behaviour in the environment. Include sources, bioaccumulation, and biomagnification with examples.

SECTION 2: AIR POLLUTION — DEFINITION & CLASSIFICATION

2.1 Definition of Air Pollution

Air pollution is the presence of one or more harmful substances — gases, particulates, or biological molecules — in the atmosphere in concentrations sufficient to cause harm to human health, living organisms, the natural environment, or the built environment. It can originate from **natural sources** (volcanic eruptions, dust storms, wildfires) or **anthropogenic sources** (industries, vehicles, burning of fossil fuels).

2.2 A. Primary Pollutants — Emitted Directly from a Source

- **Carbon Monoxide (CO):** Produced by incomplete combustion of fuel in vehicles and industries. Colourless and odourless — the 'silent killer.' Binds to haemoglobin 240x more strongly than oxygen → carboxyhaemoglobin → reduces oxygen delivery to tissues.
- **Sulfur Dioxide (SO₂):** Released by burning coal and oil in power plants, and volcanic eruptions. Irritates airways; reacts with atmospheric moisture to form sulfuric acid (H₂SO₄) — the main cause of acid rain.
- **Nitrogen Oxides (NO_x):** NO and NO₂ from high-temperature combustion in vehicle engines and thermal power plants. Precursors to photochemical smog and acid rain.
- **Particulate Matter (PM_{2.5} / PM₁₀):** Microscopic dust, smoke, and exhaust particles. PM_{2.5} (< 2.5 µm) penetrates deep into alveoli, entering the bloodstream — causes cardiovascular disease and lung cancer.
- **Volatile Organic Compounds (VOCs):** Unburned fuel vapours from vehicles and industry. React with NO_x in sunlight to form photochemical smog.
- **Lead (Pb):** From battery industries and smelters. A neurotoxin causing permanent cognitive impairment in children.

2.3 B. Secondary Pollutants — Formed by Atmospheric Reactions

- **Ground-Level Ozone (O₃):** Formed when NO_x and VOCs react in the presence of UV sunlight. Beneficial in stratosphere; toxic at ground level — damages lungs and crops.
- **Acid Rain (H₂SO₄ / HNO₃):** SO₂ / NO_x + H₂O → sulfuric / nitric acid → dissolves in rain → pH < 5.6. Damages lakes, forests, and stone buildings.
- **PAN (Peroxyacetyl Nitrate):** Formed from ozone + VOCs. A key component of photochemical smog. Causes severe eye irritation and plant damage — 10x more toxic to plants than ozone.

■ **EXAM TIP:** 7-Mark Q: Classify air pollutants into primary and secondary. Explain any three primary pollutants in detail.

SECTION 3: IMPACT OF SO_x (SULFUR OXIDES)

3.1 Sources of SO_x

- Burning of coal (1–4% sulfur content) and petroleum in thermal power plants and industries — ~50% of global SO₂ emissions
- Volcanic eruptions — major natural source (Mt. Pinatubo, 1991: 20 million tonnes SO₂ in one eruption)
- Metal smelting — roasting of sulfide ores (FeS₂, CuS) releases SO₂
- Petroleum refining and diesel vehicle exhaust

3.2 Health Effects of SO_x

■ Respiratory Effects

SO₂ is a highly soluble, acidic gas that dissolves in the moisture lining the respiratory tract, producing sulfurous acid. This irritates mucous membranes, causing **coughing, wheezing, and shortness of breath**. Even brief exposure at 0.5 ppm worsens pre-existing asthma. Prolonged exposure leads to **Chronic Obstructive Pulmonary Disease (COPD)** — an irreversible, progressive lung disease.

■ Eye & Mucous Membrane Irritation

SO₂ causes burning sensations in the eyes, lacrimation (tearing), conjunctivitis, and irritation of nasal passages. Workers in sulfur-rich industrial environments face constant exposure.

■ Cardiovascular Effects

Chronic SO₂ exposure triggers systemic inflammation and oxidative stress, increasing the risk of cardiovascular disease. Fine sulfate PM_{2.5} particles formed from SO₂ reactions further contribute to cardiovascular mortality.

3.3 Environmental Effects of SO_x

■ Acid Rain Formation

SO₂ reacts with atmospheric water vapour and oxygen to form sulfuric acid, which dissolves in rainfall producing acid rain (pH below 5.6). Acid rain: **acidifies lakes** killing fish and aquatic life; **leaches calcium and magnesium** from forest soils causing forest dieback; **corrodes stone buildings** through: $\text{H}_2\text{SO}_4 + \text{CaCO}_3 \rightarrow \text{CaSO}_4 + \text{CO}_2 + \text{H}_2\text{O}$.

■ Sulfate Aerosols

SO₂ in the atmosphere reacts to form fine sulfate aerosol particles (PM_{2.5}). These aerosols reduce visibility, contribute to lung disease, and have a temporary cooling effect on climate by reflecting incoming sunlight (global dimming).

3.4 Control of SO_x Pollution

- **Fuel Switching:** Replace high-sulfur coal with natural gas or renewable energy
- **Flue Gas Desulphurisation (FGD):** Wet scrubbers spray lime slurry (Ca(OH)₂) to absorb SO₂ → converts to gypsum (CaSO₄). Removes up to 95% of SO₂ from power plant exhaust
- **Coal Desulphurisation:** Physical and chemical treatment of coal before combustion to remove sulfur compounds
- **Low-Sulfur Fuels:** Use low-sulfur diesel; MARPOL Annex VI limits sulfur in marine fuel to 0.5%

■ **EXAM TIP:** 7-Mark Q: Explain sources and impact of SO_x on human health and environment. Describe control measures including FGD.

SECTION 4: IMPACT OF NO_x (NITROGEN OXIDES)

4.1 Sources of NO_x

- Vehicle engines (petrol and diesel) — the largest urban source of NO_x
- Thermal power plants burning coal, gas, or oil
- Industrial furnaces, cement kilns, and steel plants
- Lightning — a natural source producing NO from atmospheric N₂ and O₂
- Aircraft engines at high altitude

4.2 Health Effects of NO_x

■ Airway Irritation & Lung Inflammation

NO₂ is a reddish-brown gas with a pungent smell. It inflames the lining of the lungs, causing persistent coughing, wheezing, and chest tightness. Unlike SO₂ which affects upper airways, NO₂ penetrates **deeper into the alveoli**, causing oxidative damage to lung tissue.

■ Aggravation of Pre-existing Conditions

NO₂ worsens asthma, bronchitis, emphysema, and COPD. Children exposed to high NO_x levels show measurably **stunted lung development**, permanently reducing lung capacity and increasing susceptibility to respiratory infections.

■ Photochemical Smog

NO_x — especially NO₂ — is the key precursor to photochemical smog. In sunlight, NO₂ undergoes photolysis releasing atomic oxygen which forms ground-level ozone. (Detailed in Section 5.)

■ Acid Rain (HNO₃)

NO_x + H₂O → HNO₃ (nitric acid) → acid rain. NO_x is responsible for approximately **30% of acid rain** in most industrialised countries.

4.3 Comparison: SO_x vs NO_x

Parameter	SO _x (Sulfur Oxides)	NO _x (Nitrogen Oxides)
Primary Gas	Sulfur Dioxide (SO ₂)	Nitrogen Dioxide (NO ₂)
Main Sources	Coal burning, volcanic eruptions, smelting	Vehicles, power plants, lightning
Health Effect	Coughing, asthma, COPD	Airway inflammation, lung damage
Eye Effects	Burning sensation, conjunctivitis	Minimal direct eye irritation
Secondary Pollutant	Acid rain (H ₂ SO ₄), sulfate PM _{2.5}	Photochemical smog, ozone, HNO ₃
Gas Colour	Colourless (sharp smell)	Reddish-brown (pungent)

4.4 Control of NO_x Pollution

- **Three-Way Catalytic Converters:** Platinum/palladium/rhodium catalysts convert NO_x → N₂, CO → CO₂, HC → CO₂+H₂O. Efficiency ~90%
- **Selective Catalytic Reduction (SCR):** At power plants, ammonia injected into flue gas reacts over a catalyst: NO_x + NH₃ → N₂ + H₂O
- **BS-VI Emission Norms:** India adopted Bharat Stage VI (equivalent to Euro 6) from April 2020 — reduces NO_x from diesel vehicles by 70%
- **Low-NO_x Burners:** Staged combustion and flue gas recirculation reduce peak flame temperatures, preventing thermal NO_x formation
- **Electric Vehicles:** Zero tailpipe emissions — eliminates vehicle NO_x contribution entirely

SECTION 5: PHOTOCHEMICAL SMOG

5.1 Definition

Photochemical smog (also called '**Los Angeles smog**') is a type of **secondary air pollution** formed when nitrogen oxides (NO_x) and volatile organic compounds (VOCs) from vehicle exhaust react in the presence of **UV sunlight** to produce ozone (O₃), PAN (Peroxyacetyl Nitrate), and other oxidants. It appears as a brownish-yellow haze over cities — especially in sunny, traffic-heavy urban areas with poor air circulation.

5.2 Step-by-Step Formation Mechanism

The formation occurs through a photochemical chain reaction:

- **Step 1 — Emission:** Vehicles and industries emit NO₂ and VOCs (unburned hydrocarbons) into atmosphere during morning rush hours
- **Step 2 — Photolysis of NO₂:** UV radiation from sunlight breaks NO₂: $\text{NO}_2 + \text{UV light} \rightarrow \text{NO} + \text{O}^\bullet$ (free atomic oxygen)
- **Step 3 — Ozone Formation:** Free atomic oxygen reacts with molecular oxygen: $\text{O}^\bullet + \text{O}_2 \rightarrow \text{O}_3$ (ozone)
- **Step 4 — Reaction with VOCs:** Ozone reacts with VOCs → forms PAN, formaldehyde, aldehydes, and other oxidants
- **Step 5 — Smog:** The mixture of O₃ + PAN + aldehydes creates the brownish photochemical smog visible over cities by midday

5.3 Conditions Favouring Formation

- **Intense Sunshine:** High UV radiation intensity drives photochemical reactions — worst on clear, sunny days
- **Dense Vehicle Traffic:** Primary source of NO_x and VOC emissions — smog peaks 2–4 hours after morning rush hour
- **Low Wind Speed:** Stagnant air allows pollutants to accumulate — calm days are worst
- **Geographical Trapping:** Mountains surrounding cities (Los Angeles, Delhi, Mexico City) trap polluted air in valleys
- **Temperature Inversion:** A layer of warm air traps cooler, polluted air below — prevents normal upward dispersal of pollutants

5.4 Effects of Photochemical Smog

■ Human Health Effects

- **Eye & Throat Irritation:** Ozone and PAN cause intense burning of eyes, lacrimation, and throat inflammation. PAN is 10x more irritating than ozone to eyes
- **Respiratory Damage:** Ozone at 0.1 ppm causes chest pain, coughing, shortness of breath. Long-term exposure permanently reduces lung function
- **Aggravation of Conditions:** Worsens asthma, COPD, bronchitis — can trigger life-threatening attacks in sensitive individuals

■ Environmental & Material Effects

- **Crop Damage (Phytotoxicity):** Ozone enters plant stomata, disrupts photosynthesis, reduces crop yields by 5–15%. Most sensitive: wheat, rice, soybeans
- **Reduced Visibility:** The brownish haze scatters light, significantly reducing visibility in urban areas
- **Material Damage:** Ozone causes rubber cracking, degrades paints, textiles, and plastic materials

5.5 Control Measures

- **Electric Vehicles (EVs):** Zero tailpipe emissions — eliminates both NO_x and VOC at source
- **Catalytic Converters:** Reduce NO_x and VOC from petrol/diesel vehicles by ~90%
- **CNG & LPG Adoption:** Far less NO_x and VOC than petrol/diesel (e.g., Delhi's buses converted to CNG in 2001)
- **Strict Emission Norms (BS-VI):** Significantly reduced per-vehicle NO_x and PM since April 2020
- **Public Transport:** Metro, buses, cycling lanes reduce private vehicle numbers — reducing NO_x and VOC at source

■ **EXAM TIP:** 7-Mark Q: Explain the formation mechanism, conditions favouring formation, effects, and control of photochemical smog.

SECTION 6: INDOOR AIR POLLUTION (IAP)

6.1 Definition & Significance

Indoor Air Pollution (IAP) is the contamination of air within and around buildings due to physical, chemical, and biological pollutants from indoor activities, building materials, furnishings, and inadequate ventilation. Humans spend approximately **90% of their time indoors**. WHO estimates IAP causes **3.8 million deaths per year** globally. Indoor concentrations can be **2–5 times higher** than outdoor levels in modern tightly-sealed buildings.

6.2 Major Indoor Air Pollutants & Their Sources

■ Carbon Monoxide (CO)

Produced by incomplete combustion in gas stoves, kerosene heaters, and poorly maintained fireplaces. Colourless and odourless — the 'silent killer.' CO binds to haemoglobin, preventing oxygen delivery. At 200 ppm → headache and dizziness; at 800 ppm → unconsciousness within 2 hours; at 1600 ppm → death. Carbon monoxide detectors are essential.

■ Volatile Organic Compounds (VOCs)

Released ('off-gassed') from paints, varnishes, adhesives, cleaning products, synthetic carpets, and new furniture. Key VOCs: **benzene** (carcinogenic — from tobacco smoke, stored fuel), **formaldehyde** (irritant and carcinogen from pressed wood products). Cause eye/nose/throat irritation, headaches, and long-term liver and kidney damage.

■ Particulate Matter (PM_{2.5})

From cooking smoke (especially biomass/chulha stoves — a major problem in rural India), tobacco smoke, incense, and candles. Fine PM_{2.5} particles penetrate deep into alveoli and enter the bloodstream, causing cardiovascular disease and lung cancer. Cooking on traditional chulhas is a leading cause of IAP mortality in Indian rural women.

■ Biological Pollutants

Include mold and mildew spores (damp walls/bathrooms), dust mites (carpets, mattresses), pet dander, cockroach allergens, and bacteria/viruses. These trigger allergies, asthma attacks, and respiratory infections in sensitive individuals.

■ Radon Gas

A naturally occurring radioactive gas from uranium decay in soil, seeping through basement floors and walls. Colourless and odourless. It is the **second leading cause of lung cancer globally** (after cigarette smoking). Radioactive decay products irradiate lung tissue over years of exposure. High in regions with granite-rich geology.

■ Asbestos Fibres

Formerly used in building insulation, ceiling tiles, and pipe lagging. When disturbed during renovation, microscopic fibres are released. Chronic inhalation causes **asbestosis** (lung fibrosis) and **mesothelioma** (cancer of the pleural lining of lungs). Banned in most developed countries but still present in many older Indian buildings.

6.3 Building-Related Health Syndromes

Syndrome	Description	Cause	Resolution
Sick Building Syndrome (SBS)	Headaches, fatigue, dizziness, eye irritation — no specific illness identified	Poor ventilation, VOC off-gassing, chemical pollutants	Symptoms resolve when leaving the building
Building-Related Illness (BRI)	Diagnosable illness (e.g., Legionnaires' disease — a severe pneumonia)	Specific pathogen — Legionella bacteria in HVAC cooling towers	Symptoms persist even after leaving the building

6.4 Control of Indoor Air Pollution

- **Adequate Ventilation:** Fresh air through windows, exhaust fans, and HVAC systems with HEPA filters removes and dilutes pollutants
- **Clean Cooking Fuels:** Replace biomass chulhas with LPG, CNG, or electric induction cooktops (India's Ujjwala Scheme)
- **Low-VOC Materials:** Use low-VOC paints and adhesives; allow new furniture to off-gas outdoors before bringing inside
- **Radon Mitigation:** Sub-slab depressurisation systems draw radon from beneath foundations and vent it safely outdoors
- **Ban Indoor Smoking:** Tobacco smoke is the single largest controllable source of indoor PM2.5, CO, and carcinogenic VOCs

■ **EXAM TIP:** 7-Mark Q: Explain Indoor Air Pollution — sources, pollutants (CO, VOCs, PM, Radon, Asbestos), health effects, SBS vs BRI, and control measures.

SECTION 7: FACTORS ALTERING COMPOSITION OF THE ATMOSPHERE

7.1 Normal Atmospheric Composition

The Earth's atmosphere: N₂ (78%), O₂ (21%), Ar (0.93%), CO₂ (0.04%), and trace gases. Human activities are significantly altering this balance, driving climate change, ozone depletion, and increased air pollution.

7.2 Deforestation

Forests are the 'lungs of the Earth.' Through photosynthesis, trees absorb CO₂ and release O₂, acting as critical **carbon sinks**. The Amazon rainforest alone stores approximately 150–200 billion tonnes of carbon.

■ How Deforestation Alters the Atmosphere

- Eliminates carbon sinks — atmospheric CO₂ levels rise
- Release of stored carbon — burning trees releases centuries of stored carbon as CO₂ and CH₄
- Burning of cleared forests directly releases CO₂, CO, black carbon (soot), and PM
- Deforestation contributes ~10–15% of global greenhouse gas emissions annually
- Reduces O₂ production and disrupts regional rainfall patterns through reduced evapotranspiration

7.3 Burning of Fossil Fuels

The combustion of coal, petroleum, and natural gas is the **single largest driver of human-caused climate change**. Since the Industrial Revolution (~1760), atmospheric CO₂ has risen from ~280 ppm to over **420 ppm (2024)**. Fossil fuel combustion releases approximately **36 billion tonnes of CO₂ annually**.

■ Greenhouse Effect

CO₂, methane (CH₄), water vapour, and nitrous oxide (N₂O) are greenhouse gases that absorb outgoing infrared radiation from Earth's surface and re-emit it, warming the planet. The **enhanced greenhouse effect** due to massive increases in GHG from fossil fuels drives dangerous climate change. Global average temperature has already risen by **~1.1°C above pre-industrial levels**.

■ Pollutants Released

- **CO₂**: Primary greenhouse gas driving global warming and ocean acidification
- **SO₂**: Causes acid rain and sulfate aerosols
- **NO_x**: Precursor to photochemical smog and acid rain
- **Black Carbon (Soot)**: A potent short-lived climate forcer — absorbs solar radiation and warms atmosphere
- **Mercury (Hg)**: Released from coal combustion — bioaccumulates in aquatic food chains

7.4 Industrial Emissions

Heavy industries — steel manufacturing, cement production (calcination), chemical synthesis, and petroleum refining — emit vast quantities of CO₂, NO_x, heavy metal particles (lead, mercury, cadmium), dioxins, furans, and VOCs. Cement production alone accounts for ~8% of global CO₂ emissions. Industrial VOCs and NO_x contribute to urban smog formation.

7.5 Vehicular Emissions

With over **1.4 billion motor vehicles worldwide**, vehicular emissions are a dominant factor in urban air quality. Vehicles emit CO₂, NO_x, CO, VOCs, and PM_{2.5} from tailpipes. NO_x from vehicles reacts with atmospheric ozone, contributing to ground-level ozone (smog). Black carbon (diesel soot) is a potent short-lived climate forcer.

■ **NOTE:** India has 300+ million registered vehicles. Delhi's Air Quality Index (AQI) regularly exceeds 400 (Severe) in winter due to vehicle emissions + stubble burning + temperature inversion — a combination of multiple atmospheric factors.

7.6 Chlorofluorocarbons (CFCs) & Ozone Depletion

■ What are CFCs?

Chlorofluorocarbons (CFCs) are synthetic compounds (e.g., Freon — CCl₂F₂) historically used as refrigerants in ACs and refrigerators, aerosol spray propellants, and foam packaging agents. They are chemically inert at ground level — but this stability is what makes them so damaging in the stratosphere.

■ Ozone Depletion Mechanism

CFCs rise to the stratosphere (15–50 km altitude) and remain there for **50–100 years**. Intense UV radiation breaks CFC molecules, releasing free chlorine atoms ($\text{Cl}\cdot$). Each chlorine atom catalytically destroys up to **100,000 ozone molecules**:

- Step 1: $\text{CF}_2\text{Cl}_2 + \text{UV} \rightarrow \text{CF}_2\text{Cl}\cdot + \text{Cl}\cdot$ (CFC breakdown)
- Step 2: $\text{Cl}\cdot + \text{O}_3 \rightarrow \text{ClO}\cdot + \text{O}_2$ (ozone destroyed)
- Step 3: $\text{ClO}\cdot + \text{O} \rightarrow \text{Cl}\cdot + \text{O}_2$ (chlorine atom regenerated — cycle repeats)
- Net: $\text{O}_3 + \text{O} \rightarrow 2\text{O}_2$ (ozone permanently lost, UV-B reaches Earth's surface)

■ Effects of Ozone Depletion

- **Increased UV-B Radiation:** The ozone layer shields Earth from harmful UV-B. A thinner layer allows more UV-B to reach the surface
- **Skin Cancer & Cataracts:** UV-B causes melanoma and other skin cancers; damages the eye lens causing cataracts (leading cause of blindness)
- **Immune Suppression:** UV-B suppresses the human immune system, increasing susceptibility to infections
- **Crop Damage:** Excess UV-B reduces crop yields of legumes, rice, and soybeans by damaging DNA and photosynthetic machinery
- **Marine Ecosystem Disruption:** UV-B kills phytoplankton in surface ocean waters — disrupting the base of marine food chains

■ Montreal Protocol (1987)

The Montreal Protocol on Substances that Deplete the Ozone Layer (1987) is widely regarded as the **most successful international environmental treaty**. It phased out production and use of CFCs globally. The ozone layer is slowly recovering and is expected to fully heal by **2060–2070**. However, CFCs already in the stratosphere will continue causing damage for decades.

■ **EXAM TIP:** 7-Mark Q: Explain factors responsible for altering the composition of the atmosphere. OR: Describe the CFC ozone depletion mechanism with steps. Include the Montreal Protocol and effects of ozone depletion.

SECTION 8: TECHNIQUES TO CONTROL AIR POLLUTION

8.1 Source Reduction — Prevention at Source

- **Fuel Switching:** Replace coal/diesel with CNG, LPG, natural gas, or renewables. CNG adoption in Delhi's buses dramatically reduced air pollution after mandatory adoption in 2001.
- **Improved Combustion Efficiency:** Advanced burner designs ensure complete burning → less CO and unburned hydrocarbons. Complete combustion: $\text{fuel} + \text{O}_2 \rightarrow \text{only CO}_2 + \text{H}_2\text{O}$.
- **Catalytic Converters:** Three-way catalytic converters in vehicles use Pt/Pd/Rh catalysts: (1) CO → CO₂, (2) HC → CO₂+H₂O, (3) NO_x → N₂. Efficiency ~90%. Requires unleaded fuel.
- **Renewable Energy:** Solar, wind, hydropower generate electricity with zero direct air pollution. India targets 500 GW renewable energy by 2030.
- **BS-VI Emission Norms:** Adopted April 2020 (equivalent to Euro 6). Reduced PM from diesel by 80%, NO_x from diesel by 70% vs BS-IV.

8.2 Particulate Matter Control Devices

■ 1. Cyclone Separator

Principle — Centrifugal Force: Dust-laden air enters tangentially at the top of a cone-shaped chamber. Tangential entry creates a spinning vortex. Heavier particles are flung outward by centrifugal force → contact the wall → spiral downward into a collection hopper at the bottom. Cleaned air exits from a central vortex tube at the top (inner vortex).

- Applications: Cement plants, flour mills, grain elevators, rock crushers
- Efficiency: 70–90% for particles > 10 μm; poor for fine particles (PM_{2.5})
- Advantages: No moving parts, low maintenance, low cost, handles high temperatures
- Limitation: Ineffective for fine particles — often used as pre-cleaner before ESP or bag filter

■ 2. Electrostatic Precipitator (ESP)

Principle — Electric Charge Attraction: Dirty flue gas enters a chamber with high-voltage discharge electrodes (30,000–60,000 volts). Corona discharge ionises the gas, imparting a **negative charge** to fly ash particles. Charged particles migrate toward positively charged collector plates and adhere. Periodic mechanical rapping dislodges collected dust into hoppers below.

- Applications: Thermal power plants (primary fly ash control), cement factories, steel mills
- Efficiency: 99%+ (even for very fine particles and fly ash)

- Advantages: Handles large gas volumes, very high efficiency, low pressure drop
- Limitation: High capital cost; not effective for sticky or electrically resistive particles; high-voltage safety hazard

■ 3. Bag Filter (Fabric Filter)

Principle — Mechanical Filtration: Dust-laden gas passes through woven or felted fabric bags (glass fibre, polyester, PTFE). Particulate matter is trapped on the outer surface of the fabric, forming a dust cake that improves filtration. Bags are periodically cleaned by pulse-jet compressed air blasts — dislodged dust falls into hoppers.

- Applications: Cement, steel, pharmaceutical, food processing, municipal incinerators
- Efficiency: 99%+ for fine dust including PM2.5
- Advantages: Very high efficiency for fine particles; consistent performance; versatile
- Limitation: High pressure drop (energy cost); fabric degradation at high temperatures; bags need periodic replacement

■ 4. Wet Scrubber

Principle — Absorption in Liquid: Polluted gas enters a scrubbing chamber. Water or alkaline liquid (NaOH solution for SO₂, lime slurry for acid gases) is sprayed from nozzles, creating a fine mist. Particles collide with liquid droplets and are captured; soluble gases (SO₂, HCl, HF) are absorbed into the liquid. Cleaned gas exits from the top; contaminated slurry flows to a treatment system.

- Applications: Chemical plants, acid mist control, SO₂ removal (Flue Gas Desulphurisation)
- Efficiency: 90–99% for BOTH particles AND gaseous pollutants simultaneously
- Advantages: Removes particles + gases in one unit; cools hot gases
- Limitation: Creates liquid waste (contaminated slurry); scaling and corrosion; high water use

8.3 Comparison of Control Devices

Device	Principle	Pollutant Removed	Efficiency	Applications
Cyclone Separator	Centrifugal force	Coarse particles (>10 µm)	70–90%	Cement, flour mills
Electrostatic Precipitator	Electric charge	Fine particles, fly ash	99%+	Thermal power plants
Bag Filter	Fabric filtration	Fine particles, PM2.5	99%+	Cement, steel, pharma

Device	Principle	Pollutant Removed	Efficiency	Applications
Wet Scrubber	Liquid absorption	Particles + SO ₂ , HCl, HF	90–99%	Chemical plants, FGD
Catalytic Converter	Catalytic reaction	CO, NO _x , hydrocarbons	~90%	Automobiles

■ **EXAM TIP:** 7-Mark Q: Describe any two particulate control devices with working principle, diagram description, efficiency, and applications. OR: Compare all four air pollution control devices.

SECTION 9: AMBIENT AIR QUALITY & CONTINUOUS MONITORING

9.1 Definition

Ambient air quality refers to the condition of outdoor air with respect to concentrations of pollutants that may affect human health, ecosystems, and built structures. It is measured and regulated against **National Ambient Air Quality Standards (NAAQS)** — legally prescribed maximum permissible concentrations of specific air pollutants.

9.2 NAAQS — India (CPCB, 2009)

Set by the Central Pollution Control Board (CPCB) under the Air (Prevention and Control of Pollution) Act, 1981.

Pollutant	24-Hour Standard	Annual Standard	Key Health Effect
Sulfur Dioxide (SO ₂)	80 µg/m ³	50 µg/m ³	Respiratory disease, acid rain
Nitrogen Dioxide (NO ₂)	80 µg/m ³	40 µg/m ³	Lung inflammation, smog
PM ₁₀ (Coarse particles)	100 µg/m ³	60 µg/m ³	Respiratory, cardiovascular
PM _{2.5} (Fine particles)	60 µg/m ³	40 µg/m ³	Deep lung damage, cancer
Carbon Monoxide (CO)	2 mg/m ³ (8-hr avg)	—	Oxygen deprivation
Ozone (O ₃)	100 µg/m ³ (8-hr avg)	—	Lung damage, crop loss
Lead (Pb)	—	0.5 µg/m ³	Neurological damage

■ **NOTE:** WHO 2021 PM_{2.5} annual limit is 5 µg/m³ vs India's 40 µg/m³ — showing how far Indian cities need to improve. Delhi's annual PM_{2.5} averages 90–110 µg/m³.

9.3 Continuous Ambient Air Quality Monitoring (CAAQM)

Continuous monitoring uses automated instruments at fixed stations to measure pollutant concentrations **24 hours a day, 365 days a year**, with real-time readings every 15–60 minutes. Far richer data than old manual methods (only 24-hr average, twice a week).

■ Key Components of a CAAQM Station

- **Analyser Instruments:** Specific for each pollutant: UV fluorescence analyser (SO₂), chemiluminescence analyser (NO_x), Beta Attenuation Monitor — BAM (PM_{2.5}/PM₁₀), Non-Dispersive InfraRed — NDIR (CO and CO₂), UV photometric (ozone)
- **Meteorological Sensors:** Wind speed/direction anemometers, temperature/humidity sensors, barometric pressure — meteorological data directly affects pollutant dispersal and concentration
- **Data Logger & SCADA:** Supervisory Control and Data Acquisition systems log measurements every 15 min and transmit wirelessly to a central server for real-time public display
- **Calibration System:** Automatic zero-span calibration using certified reference gas cylinders — ensures instrument accuracy. Typically performed daily/weekly
- **Temperature-Controlled Enclosure + UPS:** Instruments housed in climate-controlled shelters (15–25°C) with Uninterruptible Power Supply — ensures accurate 24/7 operation

9.4 Air Quality Index (AQI)

A standardised index (0–500) translating pollutant concentration data from 8 pollutants (PM₁₀, PM_{2.5}, NO₂, SO₂, CO, O₃, NH₃, Pb) into a single number understandable by the public. The highest sub-index value of any pollutant becomes the overall AQI.

AQI Range	Category	Health Impact	Who is Affected
0 – 50	Good	Minimal impact	None
51 – 100	Satisfactory	Minor discomfort for sensitive people	Very sensitive individuals
101 – 200	Moderate	Breathing discomfort	Asthma, heart disease patients
201 – 300	Poor	Breathing difficulties	General public on prolonged exposure
301 – 400	Very Poor	Serious respiratory effects	Everyone — avoid outdoor activity
401 – 500	Severe/Emergency	Health emergency	Everyone severely affected

9.5 National Programmes

- **NAMP:** National Ambient Air Quality Monitoring Programme — CPCB operates 900+ stations in 344 cities. Monitors SO₂, NO_x, PM₁₀, PM_{2.5} regularly
- **NCAP 2019:** National Clean Air Programme — India's first national air quality action plan. Targets 20–30% reduction in PM_{2.5} and PM₁₀ by 2024 vs 2017 baseline in 132 non-attainment cities

■ **EXAM TIP:** 7-Mark Q: Explain Ambient Air Quality Monitoring — NAAQS standards, components of a CAAQM station, AQI with all 6 categories, and national programmes (NAMP, NCAP).

SECTION 10: KYOTO PROTOCOL & CARBON CREDITS

10.1 Kyoto Protocol

The **Kyoto Protocol** is an international treaty adopted in **Kyoto, Japan in December 1997** under the United Nations Framework Convention on Climate Change (UNFCCC). It entered into force on **16 February 2005**. It was the first international agreement to impose **legally binding GHG reduction targets** on developed (Annex I) countries.

■ Greenhouse Gases Covered

Regulated six major GHGs: CO₂, CH₄ (methane), N₂O (nitrous oxide), HFCs (hydrofluorocarbons), PFCs (perfluorocarbons), and SF₆ (sulphur hexafluoride).

■ Commitment Periods

- **First Period (2008–2012):** Developed (Annex I) countries committed to reduce collective GHG emissions by an average of 5.2% below 1990 baseline levels
- **Second Period (2013–2020):** Doha Amendment — more ambitious targets; participated primarily by EU and a few others

■ Principle of CBDR

Common But Differentiated Responsibilities (CBDR): Developed nations (Annex I) historically responsible for most GHG accumulation — bear binding targets. Developing nations (India, China, Brazil) had no binding targets under Kyoto — only voluntary actions.

■ Three Flexibility Mechanisms

- **Emissions Trading (ET):** Countries that reduce emissions below their target can SELL surplus allowances to countries above their targets — 'cap-and-trade'
- **Clean Development Mechanism (CDM):** Developed countries finance emission-reduction projects (renewable energy, reforestation) in developing countries and earn Certified Emission Reductions (CERs). Also transfers clean technology.
- **Joint Implementation (JI):** Two Annex I countries collaborate on emission-reduction projects — investing country earns Emission Reduction Units (ERUs)

■ Limitations & Successor

The **USA refused to ratify** Kyoto in 2001 (then world's largest emitter). Canada withdrew in 2011. The **Paris Agreement (2015)**, signed by 196 countries including USA, India, and China, replaced Kyoto with universal commitments from all nations — though targets are nationally determined rather than legally binding.

10.2 Carbon Credits — Cap-and-Trade System

■ Definition of a Carbon Credit

A **carbon credit** is a tradeable certificate representing the right to emit **one tonne of CO₂** (or CO₂-equivalent of other GHGs). It is the fundamental instrument of emissions trading systems. 1 carbon credit = 1 tonne CO₂.

■ How Cap-and-Trade Works

- Step 1 — CAP set: Government sets maximum total GHG emissions allowed for the year
- Step 2 — Allowances: Total cap divided into allowances (each = 1 tonne CO₂) and distributed to companies
- Step 3 — Under-cap companies: Those that emit LESS than their allocation → SELL surplus credits → earn revenue
- Step 4 — Over-cap companies: Must BUY additional allowances from market OR face large financial penalties
- Step 5 — Cap lowered annually: Overall emissions decline over time while market finds the cheapest reduction paths

■ Types of Carbon Markets

- **Compliance Markets (Mandatory):** EU Emissions Trading System (EU ETS) — 10,000+ installations across 30 countries. India's Carbon Credit Trading Scheme (CCTS, 2023) — first developing country with mandatory domestic carbon market
- **Voluntary Carbon Markets:** Companies voluntarily offset emissions. Standards: Gold Standard, Verified Carbon Standard (VCS). Airlines offering passenger carbon offsetting is an example.

■ Benefits & Criticisms

Aspect	Benefits	Criticisms
Economic	Finds cheapest global emission reduction path	Rich companies 'buy' their way out without actual reduction

Aspect	Benefits	Criticisms
Technology	CDM transfers clean technology to developing nations	CDM projects sometimes had questionable additionality
Flexibility	Countries/companies choose how and where to reduce	Complex MRV systems expensive and can be gamed
Revenue	Funds flow to green projects and reforestation	Carbon leakage — industries relocate to countries with no carbon price

■ **NOTE:** India launched CCTS in 2023 — first developing country with mandatory domestic carbon market. India's NDC: reduce emissions intensity by 45% of 2005 levels by 2030. Net-zero by 2070.

■ **EXAM TIP:** 7-Mark Q: Explain the Kyoto Protocol and Carbon Credits. Include CBDR, three flexibility mechanisms, the cap-and-trade system, and India's CCTS 2023.

SECTION 11: BIOGEOCHEMICAL CYCLES

11.0 Introduction

A **biogeochemical cycle** is the pathway by which a chemical element cycles through the biotic (living organisms) and abiotic (atmosphere, hydrosphere, lithosphere) compartments of the ecosystem. The syllabus covers four cycles: **Carbon, Nitrogen, Sulphur, and Oxygen**.

■ **NOTE:** Gaseous cycles (main reservoir = atmosphere): Carbon, Nitrogen, Oxygen. Sedimentary cycle (main reservoir = rocks/soil): Sulphur.

11.1 Carbon Cycle

■ Significance

Carbon is the backbone of all organic molecules and all life. CO₂ is the primary atmospheric form (~0.04% = 420 ppm in 2024, rising rapidly due to fossil fuel combustion). Pre-industrial level was ~280 ppm.

■ Key Processes

- **Photosynthesis (Carbon Fixation):** Plants, algae, and phytoplankton absorb CO₂ and use solar energy to produce glucose: $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{sunlight} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. Ocean phytoplankton account for 50% of all photosynthesis on Earth.
- **Respiration (Carbon Release):** All aerobic organisms break down glucose releasing CO₂: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$.
- **Decomposition:** Decomposer bacteria and fungi break down dead organic matter. Aerobic → CO₂. Anaerobic/waterlogged → methane (CH₄).
- **Combustion:** Burning of wood and fossil fuels releases stored carbon as CO₂ — ~36 billion tonnes CO₂ annually. Primary driver of anthropogenic climate change.
- **Ocean Carbon Sink:** Oceans absorb ~25% of anthropogenic CO₂ annually. $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$ (carbonic acid) → ocean acidification, threatening coral reefs.
- **Geological Sink (Fossil Fuels):** Organic matter buried over millions of years → compressed → forms coal (plants), petroleum, and natural gas. Carbon locked for geological timescales, now released in decades.

■ Human Disturbance

- Fossil fuel combustion: 36 Gt CO₂/year — releasing geological carbon stored over 300+ million years
- Deforestation: removes carbon sinks, releases stored biomass — 10–15% of GHG emissions
- Result: CO₂ risen 280 → 420 ppm → driving 1.1°C global warming, sea level rise, glacial melt

11.2 Nitrogen Cycle

■ Significance

Nitrogen is essential for amino acids (proteins), nucleic acids (DNA, RNA), chlorophyll, and ATP. N₂ makes up 78% of atmosphere but most organisms cannot use it directly — it must first be 'fixed' into usable forms.

■ 5 Key Steps in Sequence

- **1. Nitrogen Fixation:** N₂ → NH₃ (ammonia). Pathways: (a) Biological — Rhizobium (root nodules of legumes), Azotobacter (free-living in soil), cyanobacteria/BGA — enzyme: NITROGENASE. (b) Atmospheric — lightning converts N₂ → NO_x → HNO₃ → soil. (c) Industrial — Haber-Bosch process: N₂ + 3H₂ → 2NH₃ at high temperature and pressure (for fertilisers).
- **2. Nitrification:** NH₄⁺ in soil → NO₂⁻ (by Nitrosomonas bacteria) → NO₃⁻ (by Nitrobacter bacteria). Both steps require aerobic (oxygen-rich) conditions. NO₃⁻ is the main form absorbed by plant roots.
- **3. Assimilation:** Plants absorb NO₃⁻ through roots → synthesise amino acids and proteins. Animals obtain nitrogen by consuming plants or other animals.
- **4. Ammonification:** When organisms die, decomposer bacteria and fungi break down nitrogen-containing proteins → release NH₃/NH₄⁺ back into soil — making nitrogen available for nitrification again.
- **5. Denitrification:** In anaerobic (waterlogged) soil, denitrifying bacteria (Pseudomonas, Thiobacillus) convert NO₃⁻ → N₂O → N₂, returning nitrogen to the atmosphere. Completes the cycle.

■ **NOTE:** NITROGENASE enzyme catalyses biological nitrogen fixation — present ONLY in prokaryotes. It is irreversibly inhibited by oxygen — which is why Rhizobium root nodules contain LEGHAEMOGLOBIN (pink pigment) to maintain low O₂ levels and protect nitrogenase.

11.3 Sulphur Cycle

■ Significance

Sulphur is essential for life — found in amino acids cysteine and methionine, and vitamins biotin and thiamine. Critical for protein folding through disulfide bonds. The sulphur cycle is primarily **sedimentary** — main reservoir is rocks/soil/ocean sediments.

■ Key Processes

- **Weathering:** Rainfall and erosion break down sulphate rocks — gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), pyrite (FeS_2) — releasing SO_4^{2-} ions into soil water for plant uptake
- **Assimilation:** Plant roots absorb SO_4^{2-} → incorporate into cysteine and methionine amino acids. Animals obtain sulphur by consuming plants.
- **Decomposition:** Decomposer bacteria break down sulphur-containing proteins → release H_2S ('rotten egg' smell). Thiobacillus bacteria oxidise H_2S → SO_4^{2-} .
- **Volcanic Emissions:** Major natural source of atmospheric SO_2 . Mt. Pinatubo (1991): 20 million tonnes SO_2 — cooled global temperatures by $\sim 0.5^\circ\text{C}$ for 2 years (sulphate aerosol cloud reflecting sunlight).
- **Fossil Fuel Combustion:** Burning coal (1–4% sulphur) releases SO_2 — primary cause of acid rain. $\sim 50\%$ of total global SO_2 emissions.
- **Acid Rain Formation:** $\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$; $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$. Acid rain ($\text{pH} < 5.6$): acidifies lakes, leaches soil nutrients causing forest dieback, corrodes limestone buildings.

11.4 Oxygen Cycle

■ Significance

O_2 constitutes $\sim 21\%$ of atmosphere, essential for aerobic respiration. Also forms the ozone layer (O_3) in the stratosphere shielding Earth from UV-B. Ocean phytoplankton produce **50–80% of Earth's oxygen**. The oxygen cycle is intimately linked to the carbon cycle through photosynthesis and respiration.

■ Key Processes

- **Photosynthesis (Production):** Primary source of O_2 : $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. Phytoplankton produce 50–80% of Earth's O_2 .
- **Aerobic Respiration (Consumption):** All aerobic organisms: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$. O_2 consumed by global respiration approximately balances O_2 produced by photosynthesis.

- **Combustion (Consumption):** Burning fossil fuels, wood, and biomass consumes O₂ and releases CO₂.
- **Decomposition (Consumption):** Aerobic decomposers consume O₂ to break down dead organic matter → CO₂, H₂O, inorganic nutrients.
- **Rusting / Weathering (Consumption):** $\text{Iron} + \text{O}_2 + \text{H}_2\text{O} \rightarrow \text{Fe(OH)}_3$. Slow geological process removes O₂ over millions of years.
- **Photolysis (Minor Production):** UV radiation splits atmospheric water vapour: $2\text{H}_2\text{O} \rightarrow 4\text{H} + \text{O}_2$. Ancient source of O₂ before photosynthetic organisms evolved.

■ **NOTE:** Oxygen and Carbon cycles are most interconnected: photosynthesis produces O₂ and consumes CO₂ simultaneously; respiration and combustion do the opposite. Deforestation and ocean pollution disrupt BOTH cycles at once.

■ **EXAM TIP:** 7-Mark Q: Describe the Carbon Cycle OR Nitrogen Cycle with all steps in detail. OR: Describe the Sulphur Cycle with a diagram. How does fossil fuel combustion affect the sulphur cycle and cause acid rain?

SECTION 12: EXAM QUESTIONS — UNIT 1

Part A — 2-Mark Questions

1. Define air pollution. Give two examples each of primary and secondary pollutants.
2. What is photochemical smog? Name the two chemical precursors that form it.
3. Define bioaccumulation and biomagnification. Give one example of each.
4. What is Non-Point Source Pollution? How is it different from Point Source?
5. Name any two sources of SO_x and state two health effects.
6. What is the role of Rhizobium in the nitrogen cycle? What enzyme does it use?
7. Define the Kyoto Protocol. When was it adopted and when did it enter into force?
8. What is a carbon credit? What does one carbon credit represent?
9. How do CFCs deplete stratospheric ozone? Name the treaty that banned CFCs.
10. Name the four air pollution control devices and state which is most efficient.
11. What is the Air Quality Index (AQI)? Name all six AQI categories.
12. What is nitrogenase? Why does it require low oxygen conditions?
13. State any two factors responsible for altering the composition of the atmosphere.
14. What is temperature inversion and how does it worsen photochemical smog?

Part B — 7-Mark Questions (OR Type)

■ **Q1(a): Explain photochemical smog — define, give its step-by-step formation mechanism, conditions favouring it, all effects (health + environmental), and control measures.**

Answer Guide: Define (Los Angeles smog, secondary pollutant) → Formation: 5 steps (NO_x + VOCs → UV → photolysis of NO₂ → O• → O₃ → PAN + smog) → 5 conditions: sunshine, traffic, low wind, temperature inversion, mountains → Health effects: eye/throat irritation, respiratory damage → Environmental: crop damage (ozone phytotoxicity), visibility reduction, rubber cracking → Control: EVs, catalytic converters, BS-VI, CNG, public transport.

■ **Q1(b) [OR]: Describe the Nitrogen Cycle with a neat diagram. Explain all 5 steps in detail with the bacteria involved.**

Answer Guide: Significance (amino acids, DNA, RNA) → N₂ = 78% but unusable directly → 5 steps: (1) Fixation — Rhizobium, Azotobacter, lightning, Haber-Bosch, enzyme nitrogenase (2) Nitrification — Nitrosomonas (NH₄⁺ → NO₂⁻), Nitrobacter (NO₂⁻ → NO₃⁻) (3) Assimilation — plants absorb NO₃⁻ (4) Ammonification — decomposers → NH₃ (5) Denitrification — Pseudomonas → N₂. Mention nitrogenase + leghaemoglobin.

■ **Q2(a): Describe all techniques for controlling air pollution. Explain any two particulate control devices with working principle, applications, and efficiency.**

Answer Guide: Source reduction: fuel switching, BS-VI, catalytic converter, renewables → Devices: ESP (electric charge, 99%+, power plants — explain corona discharge, collector plates, rapping) + Wet Scrubber (liquid absorption, 90–99%, SO₂ removal — alkaline spray, removes particles AND gases) → Include comparison table of all 4 devices.

■ **Q2(b) [OR]: Explain factors responsible for altering the composition of the atmosphere. Include the greenhouse effect and the CFC ozone depletion mechanism.**

Answer Guide: Normal composition (N₂ 78%, O₂ 21%, CO₂ 0.04%) → 5 factors: (1) Deforestation — CO₂ sink loss, 10–15% GHG (2) Fossil fuels — 36 Gt CO₂/yr, 280→420 ppm (3) Industrial emissions — metals, VOCs, dioxins (4) Vehicular — NO_x, VOCs, black carbon (5) CFCs — chain reaction: Cl• + O₃ → ClO• + O₂ → Cl• (100,000 O₃ per Cl atom) → effects: UV-B, skin cancer → Montreal Protocol 1987. Greenhouse effect: CO₂ traps IR radiation.

■ **Q3(a): Explain Indoor Air Pollution — sources, major pollutants, health effects, building syndromes (SBS and BRI), and control measures.**

Answer Guide: 90% time indoors, 3.8 million deaths/yr (WHO), 2–5x higher than outdoor → 6 pollutants: CO (gas stoves, 200 ppm→headache, 1600 ppm→death), VOCs (paints, benzene/formaldehyde), PM_{2.5} (chulha, lung cancer), biological (mold, dust mites), Radon (2nd leading lung cancer cause), Asbestos (mesothelioma) → SBS (no specific illness, resolves when leaving) vs BRI (diagnosable, Legionnaires' disease) → Control: ventilation, clean fuels, low-VOC materials.

■ **Q3(b) [OR]: Explain Ambient Air Quality Monitoring — NAAQS standards, components of a CAAQM station, and the AQI system.**

Answer Guide: Define ambient air quality → NAAQS by CPCB (2009) under Air Act 1981 — table: SO₂ 80 µg/m³, NO₂ 80 µg/m³, PM₁₀ 100 µg/m³, PM_{2.5} 60 µg/m³, CO 2 mg/m³, O₃ 100 µg/m³ → CAAQM: 5 components (UV fluorescence for SO₂, chemiluminescence for NO_x, BAM for PM, NDIR for CO, meteorological sensors, data logger, calibration, UPS) → AQI 0–500 in 6 categories → NAMP (900+ stations, 344 cities) → NCAP 2019 (20–30% PM reduction target).

■ **Q4(a): Explain the Kyoto Protocol and Carbon Credits. Include CBDR, three flexibility mechanisms, and the cap-and-trade system.**

Answer Guide: Kyoto (adopted 1997, Kyoto Japan, in force Feb 2005) → 6 GHGs covered → CBDR principle (Annex I binding, developing nations voluntary) → 2 commitment periods (2008-12: 5.2% reduction; 2013-20: Doha) → 3 mechanisms: ET (cap-and-trade), CDM (developed → developing, earn CERs), JI (Annex I collaboration, earn ERUs) → USA non-ratification 2001 → Paris Agreement 2015 successor → Carbon credit = 1 tonne CO₂ → Cap-and-trade 5 steps → India CCTS 2023.

■ **Q4(b) [OR]: Describe the Carbon Cycle with a neat diagram. How are human activities disturbing the carbon cycle and causing climate change?**

Answer Guide: Significance (backbone of all life, CO₂ main atmospheric form) → 6 processes: Photosynthesis ($6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + \text{O}_2$), Respiration, Decomposition (aerobic=CO₂, anaerobic=CH₄), Combustion, Ocean sink (25% CO₂ absorbed, acidification), Geological (fossil fuels = millions of years stored carbon) → Human disturbance: (1) Fossil fuels 36 Gt CO₂/yr (2) Deforestation 10-15% (3) Agriculture (CH₄ from rice paddies/cattle, N₂O from fertilisers) → CO₂ 280→420 ppm → 1.1°C warming → glacial melt, sea level rise, ocean acidification.

■ **Q5(a): Explain all four biogeochemical cycles — Carbon, Nitrogen, Sulphur, and Oxygen — highlighting the key processes and organisms involved in each.**

Answer Guide: Carbon: photosynthesis→respiration→decomposition→combustion→ocean sink→geological. Nitrogen: 5 steps with bacteria (Rhizobium, Nitrosomonas, Nitrobacter, Pseudomonas) + nitrogenase enzyme. Sulphur: sedimentary cycle, weathering→assimilation→decomposition→volcanic→combustion→acid rain (H₂SO₄). Oxygen: photosynthesis (phytoplankton 50-80%), respiration, combustion, decomposition, rusting — link to carbon cycle.

■ **Q5(b) [OR]: Explain the Sulphur Cycle. How does fossil fuel combustion disturb it and cause acid rain? Include control measures.**

Answer Guide: Significance (cysteine, methionine amino acids) → Sedimentary cycle, main reservoir rocks/soil → Steps: weathering (CaSO₄, FeS₂) → SO₄²⁻ → assimilation → decomposition (H₂S) → Thiobacillus (H₂S→SO₄²⁻) → volcanic emissions (Pinatubo: 20 Mt SO₂) → fossil fuel combustion (50% global SO₂) → acid rain (SO₂+H₂O→H₂SO₄, pH<5.6) → effects: lakes acidified, forest dieback, building corrosion → Control: FGD, low-sulphur fuel, coal desulphurisation.

END OF UNIT 1 NOTES | Environmental Science BHM4T02 | RTMNU NEP 2020 | B.Tech CSE (Data Science)

UNIT 2

Water Pollution & Its Control Techniques

Environmental Science | Detailed Study Notes

■ 6 Credits

■ 7-Mark Questions Included

■ Full Syllabus Covered

1. Introduction to Water Pollution

Water is the most precious natural resource on Earth. However, human activities, industrialisation, agricultural practices, and urbanisation have severely degraded water quality worldwide. **Water pollution** is defined as the introduction of undesirable substances — physical, chemical, or biological — into a water body that renders it unfit for its intended use (drinking, irrigation, aquatic life, recreation).

Definition: Water pollution occurs when harmful substances (pollutants) contaminate streams, rivers, lakes, oceans, groundwater, or other water bodies, degrading water quality and making it toxic for humans and the environment.

Types of Water Pollutants:

- **Physical pollutants:** Suspended solids, colour, turbidity, heat (thermal pollution)
- **Chemical pollutants:** Heavy metals, pesticides, fertilizers, acids, organic compounds
- **Biological pollutants:** Pathogens — bacteria, viruses, protozoa, parasites
- **Radiological pollutants:** Radioactive isotopes from nuclear plants or mining

2. Major Sources of Water Pollution

2.1 Eutrophication

Eutrophication is the process by which a water body becomes overly enriched with nutrients (primarily nitrogen and phosphorus), stimulating excessive growth of algae and aquatic plants. It is one of the most serious water quality problems globally.

Causes of Eutrophication:

- Agricultural run-off carrying nitrogen (NO_3^-) and phosphate (PO_4^{3-}) from fertilizers
- Discharge of municipal sewage and detergents rich in phosphorus
- Animal wastes from livestock farming entering water bodies
- Industrial effluents containing organic matter
- Urban storm-water run-off

Process and Effects:

- Excess nutrients → rapid algal growth → **Algal Bloom** (green/blue-green scum on water surface)
- Algal bloom blocks sunlight → aquatic plants below die
- Dead organic matter decomposed by bacteria → **Biological Oxygen Demand (BOD)** rises sharply
- **Hypoxia / Anoxia:** Dissolved oxygen (DO) drops below 2 mg/L → fish kills
- Release of toxins by cyanobacteria (blue-green algae) — harmful to humans and animals

- Loss of biodiversity; lake/pond eventually becomes a "dead zone"

Key Terms: Oligotrophic (nutrient-poor, clear) → Mesotrophic → Eutrophic → Hypertrophic (severely polluted). Cultural eutrophication is human-accelerated eutrophication.

2.2 Acid Mine Drainage (AMD)

Acid Mine Drainage (AMD) is formed when iron sulphide minerals (especially pyrite — FeS_2) in rock, soil, or mine tailings are exposed to air and water. Chemical oxidation and microbial activity produce sulphuric acid, which leaches into streams, rivers, and groundwater.

Chemical Reaction (Pyrite Oxidation):

$$\text{FeS}_2 + \text{H}_2\text{O} + \text{O}_2 \rightarrow \text{Fe}^{2+} + \text{SO}_4^{2-} + \text{H}^+ \text{ (Acid)}$$

Fe^{2+} is further oxidised to Fe^{3+} (orange/red precipitate — "yellow boy") by bacteria like *Acidithiobacillus ferrooxidans*.

Effects of AMD:

- pH of water drops to 2–4 (extremely acidic) — lethal to most aquatic organisms
- Mobilises heavy metals: arsenic (As), lead (Pb), cadmium (Cd), mercury (Hg), zinc (Zn)
- Orange/red staining of stream beds; characteristic metallic odour
- Destruction of benthic (bottom-dwelling) organisms and fish populations
- Contamination of drinking water sources and agricultural land
- Persists for decades or centuries even after mine closure

Control Measures for AMD:

- **Active treatment:** Lime dosing to neutralise acid; chemical precipitation of metals
- **Passive treatment:** Constructed wetlands, anoxic limestone drains, permeable reactive barriers
- **Prevention:** Backfilling mines, surface sealing to exclude oxygen and water

2.3 Pesticides and Fertilizers

Modern agriculture relies heavily on synthetic pesticides and fertilizers. When applied in excess or washed by rain, these chemicals enter water bodies causing serious pollution.

Pesticide Pollution:

- Categories: Organochlorines (DDT, Aldrin, Endrin), Organophosphates (Malathion, Parathion), Carbamates, Pyrethroids
- **Bioaccumulation:** Pesticides are fat-soluble; concentrated in body fat of organisms
- **Biomagnification:** Concentration increases at each trophic level (up to 1000× in top predators)

- DDT → eggshell thinning in birds; disrupts endocrine system; carcinogenic in mammals
- Runoff of pesticides → kills non-target aquatic insects, crustaceans, and fish
- Leaching to groundwater renders wells unfit for drinking

Fertilizer Pollution:

- **Nitrate leaching:** NO_3^- enters groundwater; causes **Methemoglobinemia (Blue Baby Syndrome)** in infants
- Excess phosphate → eutrophication (as discussed above)
- Ammonia from urea fertilizers is directly toxic to fish at low concentrations
- Fluoride impurities in phosphate fertilizers accumulate in soil and water

2.4 Dyeing and Tanning Industries

The textile dyeing and leather tanning industries are among the most polluting industries because of the large volumes of chemically-loaded effluents they discharge.

Dyeing Industry Pollutants:

- Synthetic dyes: Azo dyes, reactive dyes, vat dyes — highly coloured, toxic, and resistant to biodegradation
- Auxiliaries: Salts (NaCl), surfactants, fixing agents, heavy metals (chromium in mordant dyeing)
- Effluent characteristics: High BOD, COD, TDS, colour, pH variations
- Azo dye degradation produces **aromatic amines** — potent carcinogens
- Coloured water blocks sunlight penetration → affects photosynthesis in aquatic plants

Tanning Industry Pollutants:

- **Chrome tanning (most common):** Uses hexavalent chromium Cr(VI) — carcinogenic and mutagenic
- Effluent contains: sulphides, lime, salts, ammonium compounds, organic nitrogen
- High suspended solids, BOD (800–3000 mg/L), TDS, and strong odour
- **Chromium toxicity:** Damages kidneys, liver; causes skin ulcers and lung cancer
- Untreated tannery effluent destroys aquatic biodiversity completely

2. Major Sources (continued)

2.5 Marine Pollution

The oceans cover 71% of Earth's surface and are the ultimate sink for most water pollutants. Marine pollution threatens marine biodiversity, fisheries, and coastal communities worldwide.

Major Sources of Marine Pollution:

- **Oil spills:** Tanker accidents, offshore drilling blow-outs, pipeline leaks — form slicks that coat marine organisms
- **Plastic debris:** ~ 8 million tonnes of plastic enter oceans annually
- **Heavy metals:** Mercury, lead, cadmium — bioaccumulate in marine food chains
- **Sewage and nutrients:** Cause coastal eutrophication and "dead zones" (e.g., Gulf of Mexico)
- **Radioactive waste:** Historical dumping of nuclear waste containers
- **Noise pollution:** Shipping and sonar affect whale communication and navigation
- **Thermal pollution:** Cooling water from coastal power plants raises sea temperature

Effects of Oil Spills:

- Smothers marine birds and mammals; impairs thermoregulation by coating feathers/fur
- Toxic PAHs (polycyclic aromatic hydrocarbons) damage gills, livers, and reproduction of fish
- Coral reef bleaching and mortality
- Destroys mangroves and intertidal ecosystems
- Economic losses to fishing and tourism industries

2.6 Micro-plastics

Micro-plastics are plastic particles **smaller than 5 mm** in size. They have emerged as one of the most pervasive and alarming pollutants of the 21st century, found everywhere from ocean trenches to Arctic ice to human blood.

Types and Sources:

- **Primary micro-plastics:** Manufactured directly at micro-scale — microbeads in cosmetics/personal care products, plastic pellets (nurdles)
- **Secondary micro-plastics:** Result from breakdown of larger plastic items through UV, mechanical abrasion, and weathering
- **Synthetic fibres:** Released during washing of polyester/nylon clothing (up to 700,000 fibres per wash)
- **Tyre wear particles:** Major source — enter stormwater drains

- **Agricultural films:** Degrading plastic mulch films fragment in soil

Effects of Micro-plastics:

- Ingested by fish, seabirds, marine mammals, and filter-feeders (mussels, oysters)
- Physical blockage of digestive systems; false sense of satiety → starvation
- Carry and concentrate persistent organic pollutants (PCBs, DDT) on their surface — "Trojan Horse effect"
- Enter human food chain via seafood, drinking water, and even table salt
- Nano-plastics (<100 nm) can cross cell membranes and blood-brain barrier
- Endocrine disruption due to plastic additives (phthalates, bisphenol-A)

Did you know? A recent study found micro-plastics in human placenta, breast milk, and lungs. The average person ingests approximately 5 grams of plastic per week — equivalent to a credit card.

3. Techniques to Control Water Pollution

Water pollution control involves conventional and advanced treatment technologies to remove pollutants from wastewater before it is discharged into the environment or reused.

3.1 Conventional Wastewater Treatment — Types of Sewage

Sewage is the liquid waste from households, commercial establishments, and some industries. It contains 99.9% water and 0.1% impurities, yet this small fraction causes enormous environmental damage.

Types of Sewage:

- **Domestic/Sanitary sewage:** From households — contains organic matter, pathogens, nutrients (N, P), detergents
- **Industrial sewage/effluents:** From factories — may contain heavy metals, toxic chemicals, high BOD/COD
- **Stormwater/Surface runoff:** Rain-driven runoff from roads, roofs — carries silt, oil, pesticides
- **Combined sewage:** Mixture of domestic and stormwater in older piped systems

3.2 Sewerage System

A **sewerage system** is the network of pipes, pumping stations, and infrastructure that collects and transports sewage from its source to treatment facilities.

Components of Sewerage System:

- **House sewer:** Connects individual buildings to the public sewer network
- **Lateral/branch sewers:** Smaller pipes collecting from groups of buildings
- **Main sewers/trunk sewers:** Large pipes carrying sewage to interceptor sewers
- **Interceptor sewers:** Large conduits running parallel to rivers, intercepting outfalls
- **Pumping/lift stations:** Raise sewage to higher elevations when gravity flow is insufficient
- **Sewage treatment plant (STP):** Final destination for treatment

3.3 Alternative Systems

- **Septic tanks:** On-site treatment for areas not connected to sewers — anaerobic digestion of solids
- **Constructed wetlands:** Engineered wetland systems using plants and microbes — low cost, eco-friendly
- **Waste stabilisation ponds:** Large shallow lagoons using sunlight and algae for treatment
- **Package treatment plants:** Compact, pre-fabricated units for small communities or resorts

- **Biodigesters / biogas plants:** Anaerobic treatment producing biogas (CH_4) as energy

4. Wastewater Treatment — Primary, Secondary & Tertiary

4.1 Primary Treatment (Physical Treatment)

Primary treatment uses physical processes to remove large and settleable solids from wastewater. It typically removes 50–65% of suspended solids and 30–40% of BOD.

Unit Operations in Primary Treatment:

- **Screening:** Bar screens or fine screens remove large objects (rags, sticks, plastics). Coarse screens: 50–150 mm openings; Fine screens: 3–6 mm
- **Grit removal:** Grit chambers slow velocity to 0.3 m/s; sand, gravel, and eggshells settle
- **Primary sedimentation (clarifiers):** Wastewater held 1.5–2 hours; organic solids settle as **primary sludge**; grease and oil float and are skimmed off
- **Flotation:** Dissolved air flotation (DAF) for light solids and fats
- **Equalization:** Balances flow rates and pollutant loads before further treatment

Primary Sludge collected from clarifiers is typically 3–8% solids. It must be further processed through sludge digestion, thickening, dewatering, and disposal.

4.2 Secondary Treatment (Biological Treatment)

Secondary treatment uses biological processes — aerobic or anaerobic — to remove dissolved and colloidal organic matter. It reduces BOD by 85–95% and removes 90% of suspended solids.

Aerobic Processes:

- **Activated Sludge Process (ASP):** Wastewater mixed with micro-organism-rich "activated sludge" in aeration tanks. Air blown in; bacteria oxidise organic matter. Effluent clarified; sludge recycled. Most widely used secondary treatment worldwide.
- **Trickling Filter (Biofilter):** Wastewater sprinkled over a bed of stones/plastic media. Biofilm of bacteria on media degrades organics. Simple, low energy, but less efficient than ASP.
- **Rotating Biological Contactor (RBC):** Plastic discs half-submerged, rotate slowly. Biofilm alternately exposed to wastewater and air. Compact and energy-efficient.
- **Oxidation Ponds / Lagoons:** Shallow basins (0.3–1.5 m deep); sunlight, wind, and algae support aerobic bacteria. Low-cost, suitable for tropical countries.
- **Sequential Batch Reactor (SBR):** Fill–React–Settle–Decant cycle in a single tank. Flexible and suitable for small flows.

Anaerobic Processes:

Anaerobic treatment occurs **without oxygen**. Used for high-strength industrial wastewater and sludge treatment. Produces biogas (60–70% CH₄) as a valuable by-product.

- **Anaerobic Digestion:** Four stages — (1) **Hydrolysis:** complex organics broken to monomers; (2) **Acidogenesis:** monomers → volatile fatty acids + H₂ + CO₂; (3) **Acetogenesis:** fatty acids → acetate; (4) **Methanogenesis:** acetate + H₂ → CH₄ + CO₂
- **Upflow Anaerobic Sludge Blanket (UASB) Reactor:** Wastewater flows upward through a sludge blanket. Very compact; widely used in India for textile and sugar factory effluents.
- **Anaerobic Filter:** Fixed-film anaerobic reactor with inert media for biofilm attachment.
- **Imhoff Tank:** Two-chamber tank — upper settling compartment + lower digestion compartment. Traditional design, suitable for small communities.

Parameter	Aerobic Treatment	Anaerobic Treatment
Oxygen requirement	Requires oxygen (aeration)	No oxygen — strict anaerobes
Energy	Energy intensive (aeration costs)	Low energy; produces biogas
BOD removal	85–95%	50–85%
Sludge produced	High quantity	Low quantity
By-products	CO ₂ , water, cell biomass	Biogas (CH ₄ + CO ₂), digestate
Start-up time	1–2 weeks	2–4 months
Application	Municipal sewage, low-strength waste	High-strength industrial waste

4.3 Tertiary Treatment (Advanced Treatment)

Tertiary treatment further polishes the secondary effluent to produce high-quality water suitable for reuse, discharge into sensitive water bodies, or irrigation. It can remove remaining nutrients, pathogens, and trace contaminants.

- **Nutrient removal (Nitrogen):** Nitrification ($\text{NH}_4^+ \rightarrow \text{NO}_3^-$) followed by Denitrification ($\text{NO}_3^- \rightarrow \text{N}_2$ gas) using bacteria in anoxic zones
- **Phosphorus removal:** Chemical precipitation using alum ($\text{Al}_2(\text{SO}_4)_3$) or lime; or biological enhanced phosphorus removal (EBPR) in activated sludge
- **Filtration:** Rapid sand filters, multimedia filters, or membrane filters (MF, UF, NF, RO) to remove fine suspended particles and pathogens
- **Disinfection:** Chlorination (most common), UV irradiation (chemical-free), ozonation (powerful oxidiser), or chloramine treatment for pathogen destruction
- **Activated carbon adsorption:** Granular activated carbon (GAC) removes trace organics, pharmaceuticals, pesticides, and colour
- **Reverse osmosis (RO):** Semi-permeable membranes remove dissolved salts, heavy metals, nitrates — used for water reuse and desalination
- **Advanced oxidation processes (AOPs):** H_2O_2 /UV, Fenton's reagent — destroy recalcitrant micro-pollutants

Reclaimed Water Quality: After tertiary treatment, water meets standards for agricultural irrigation (Class A), industrial cooling, groundwater recharge, and in some countries, indirect potable reuse.

4.4 Safe Disposal and Its Utility

Proper disposal of treated wastewater and sludge is critical to prevent secondary pollution.

Methods of Safe Disposal:

- **Discharge to surface water:** Treated effluent must meet CPCB/SPCB standards (BOD <30 mg/L, SS <100 mg/L, pH 5.5–9)
- **Land disposal (irrigation):** Treated wastewater used for agriculture (rich in N, P); soil acts as additional treatment medium — "living filter"
- **Sludge utilization:** Dried/composted sludge (biosolids) used as soil amendment/fertilizer
- **Biogas utilization:** Methane from digesters used for electricity, cooking, or heat
- **Effluent reuse:** In industries for cooling towers, boilers (water conservation)
- **Ocean outfall:** Treated effluent discharged via submarine outfall pipes — requires EIA approval

5. Treatment Schemes for Industrial Wastewater

5.1 Dairy Industry Wastewater

Dairy wastewater originates from milk processing, cheese and butter making, yoghurt production, and cleaning of equipment. It has **very high BOD (1,000–5,000 mg/L)** due to milk fats, proteins, and lactose.

Characteristics:

- High BOD and COD from milk solids; milky/white appearance
- High fat content (emulsified milk fat)
- Fluctuating pH from acid cleaning (pH 2) and alkaline CIP (pH 12)
- High temperature effluent from pasteurisation
- Contains detergents and sanitisers (chlorine compounds)

Treatment Scheme:

- **Step 1 — Screening:** Remove coarse solids
- **Step 2 — Equalization tank:** Balance flow and pH fluctuations
- **Step 3 — DAF (Dissolved Air Flotation):** Remove emulsified fats and suspended solids
- **Step 4 — Aerobic biological treatment:** Activated sludge or trickling filter for BOD reduction
- **Step 5 — Secondary clarifier:** Separate biomass from treated effluent
- **Step 6 — Anaerobic digestion of sludge:** Produce biogas; recover energy
- **Step 7 — Polishing / Disinfection:** Chlorination before discharge

5.2 Textile Industry Wastewater

Textile effluents are among the most complex and environmentally damaging industrial wastewaters. They are **highly coloured** (from dyes), alkaline, and contain a wide array of toxic chemicals.

Characteristics:

- Strong colour (absorbs light) — impairs aquatic photosynthesis even at very low concentrations
- High BOD (200–600 mg/L), very high COD (400–3000 mg/L)
- High TDS and conductivity (due to salts used in dyeing)
- Alkaline pH (8–12); high temperature
- Heavy metals from metallic dyes (chromium, copper)
- Toxic surfactants and auxiliary chemicals

Treatment Scheme:

- **Primary:** Screening → Equalization → Coagulation-flocculation (alum/ FeCl_3) → Sedimentation — removes colour and SS
- **Secondary:** Biological treatment — ASP or UASB + aerobic (for high BOD reduction)
- **Tertiary:** Activated carbon adsorption (for residual colour and organics) → Ozonation/UV (colour destruction) → Reverse osmosis (for zero liquid discharge)
- **ZLD (Zero Liquid Discharge):** Many Indian states mandate ZLD for textile mills — all water recovered and recycled; only solid waste remains

5.3 Power Plant Wastewater

Thermal power plants (coal, gas, nuclear) generate multiple streams of wastewater from cooling, ash handling, flue gas desulfurization, and boiler operations.

- **Cooling water:** High temperature (thermal pollution); biocides (Cl_2) added to prevent biological fouling
- **Ash pond water:** Contains fly ash slurry — high suspended solids, heavy metals (As, Hg, Se), fluoride
- **FGD wastewater:** Flue gas desulphurisation effluent — high selenium, mercury, nitrates
- **Boiler blowdown:** High TDS, dissolved minerals, traces of hydrazine and morpholine

Treatment:

- Cooling water: Cooling towers (reduce temperature); biocide treatment; drift eliminator
- Ash pond: Settling basins → coagulation → sludge dewatering; fly ash utilised in cement/bricks
- FGD water: Chemical precipitation (lime) → membrane treatment → evaporation for ZLD

5.4 Pharmaceutical Industry Wastewater

Pharmaceutical effluents are characterised by the presence of **Active Pharmaceutical Ingredients (APIs), solvents, and intermediates** that are toxic, recalcitrant (not easily biodegraded), and potentially carcinogenic or mutagenic.

- Very high BOD (up to 50,000 mg/L for some fermentation streams) and COD
- Presence of antibiotics — inhibit biological treatment bacteria if untreated
- Residual antibiotics in water bodies promote antibiotic-resistant bacteria (AMR — major global health threat)
- Solvents: methanol, acetone, chlorinated solvents — toxic and volatile
- Heavy metals from catalyst and reagent usage

Treatment Scheme:

- **Solvent recovery:** Distillation to recover and reuse solvents before wastewater treatment
- **Physico-chemical treatment:** pH adjustment → coagulation-flocculation → sedimentation

- **Advanced oxidation (AOP):** Ozone, H_2O_2 /UV, or Fenton process to destroy APIs and solvents
- **Biological treatment:** High-rate anaerobic (UASB) + aerobic (ASP) — only after AOP pre-treatment
- **Activated carbon polishing:** Remove residual micro-pollutants
- **RO + Evaporation:** For ZLD compliance

5.5 Agro-Based Industries — Rice Mills

Rice mills produce wastewater primarily from parboiling (soaking and steaming of paddy before milling). This is called **parboiling effluent** and is one of the most strongly polluting agro-industrial wastewaters.

Characteristics of Rice Mill Wastewater:

- Extremely high BOD (5,000–12,000 mg/L) and COD — due to starch, proteins, sugars from rice leaching
- Dark brown colour from parboiling
- High suspended solids (rice husk fragments, bran particles)
- Strong odour from anaerobic decomposition
- Slightly acidic pH (5–6.5)

Treatment Scheme:

- **Screening:** Remove rice husk and fibrous solids
- **Settling / Primary clarifier:** Remove heavy suspended solids (starch granules)
- **Anaerobic digestion (primary biological):** UASB reactor or covered lagoon — most suitable for this high-BOD waste; produces biogas used for parboiling steam generation
- **Aerobic polishing:** Activated sludge or oxidation pond for remaining BOD
- **Constructed wetlands:** Final polishing — low-cost, eco-friendly
- **Resource recovery:** Rice husk → silica / ash for cement; biogas → energy; rice bran → oil extraction

Zero-Waste Approach: Modern rice mills aim for a zero-waste concept: parboiling effluent → biogas (energy) → digestate (fertilizer); rice husk → fuel + silica; rice bran → edible oil.

6. Important 7-Mark / Long-Answer Questions

Exam Tip: 7-mark answers should be approximately 300–500 words. Include definitions, mechanisms, a diagram description (if applicable), examples, and at least one comparison table where relevant.

Q1. What is eutrophication? Explain its causes, process, and effects on aquatic ecosystems with a suitable diagram.

Introduction: Eutrophication is the process by which a water body becomes enriched with inorganic nutrients — especially nitrogen (N) and phosphorus (P) — leading to excessive growth of algae and aquatic plants. It is a natural process but human activities (cultural eutrophication) have dramatically accelerated it.

Causes: (1) Agricultural run-off containing nitrate and phosphate fertilizers. (2) Discharge of municipal sewage and phosphate-rich detergents. (3) Animal waste from livestock entering surface water. (4) Industrial effluents containing organic nitrogen and phosphorus. (5) Urban storm-water carrying organic material from streets and lawns.

Process: The sequence is as follows: *Nutrient input* → *Algal bloom formation* (rapid growth of surface algae/cyanobacteria) → *Sunlight blockage* (sub-surface plants die) → *Decomposition by bacteria* (huge oxygen demand) → *Dissolved oxygen (DO) depletion (Hypoxia)* → *Fish kills and biodiversity loss*.

Effects: (a) Depletion of dissolved oxygen below 2 mg/L — fish suffocation. (b) Release of toxins by cyanobacteria (microcystin) — harmful to humans and livestock. (c) Foul odour and unaesthetic appearance — loss of recreational value. (d) Shift to algae-dominated ecosystem; loss of submerged aquatic vegetation. (e) Economic losses — reduced fish catch, contaminated drinking water sources. (f) Long-term: the water body may become shallow and eventually turns into a swampy land.

Control Measures: Reduce nutrient inputs by adopting precision fertilization; upgrading wastewater treatment plants to include nutrient removal (nitrification-denitrification for N; chemical phosphorus precipitation with alum); banning phosphate-based detergents; restoring riparian buffer strips along waterways.

Q2. Describe in detail the primary, secondary, and tertiary stages of wastewater treatment with appropriate diagrams and the significance of each stage.

Introduction: Municipal wastewater treatment is carried out in three progressively advanced stages, each targeting specific classes of pollutants.

Primary Treatment (Physical): Employs physical operations. (1) Screening: bar screens remove large debris; (2) Grit chamber: removes sand and gravel at controlled flow velocity (0.3 m/s); (3) Primary settling tank (clarifier): detention time 1.5–2 hours — 50–65% suspended solids settle as primary sludge; floating grease/oil is mechanically skimmed. Efficiency: removes 30–40% BOD and 50–65% SS.

Secondary Treatment (Biological): Uses microorganisms to oxidise dissolved organics. Aerobic processes: Activated Sludge Process (ASP) — aeration tank + secondary clarifier; Trickling filter — biofilm on media; Rotating biological contactor (RBC). Anaerobic processes: UASB reactor, anaerobic filter — for high-strength wastes. Efficiency: removes 85–95% BOD and 90% SS. Secondary sludge is generated.

Tertiary Treatment (Advanced): Polishes secondary effluent. Processes: Nutrient removal (nitrification + denitrification; phosphorus precipitation); Filtration (sand filters, membrane filtration); Disinfection (chlorination, UV, ozone); Activated carbon adsorption; Reverse osmosis. Produces near-potable quality water suitable for irrigation, industrial reuse, or groundwater recharge.

Significance: A well-designed three-stage system can reduce BOD from ~300 mg/L (raw sewage) to <5 mg/L (tertiary treated), meeting stringent standards. It protects receiving water bodies, prevents eutrophication, eliminates pathogens, and increasingly enables water recycling in water-scarce regions.

Q3. Write detailed notes on the treatment of textile industry wastewater. Why is Zero Liquid Discharge (ZLD) important?

Characteristics of Textile Effluent: Textile wastewater contains synthetic dyes (azo, reactive, vat dyes), large quantities of sodium chloride (salt used in dyeing), surfactants, alkaline pH (8–12), high temperature, high BOD (200–600 mg/L), very high COD (400–3000 mg/L), and heavy metals from metallic dyes. Azo dyes breakdown to produce aromatic amines — carcinogenic.

Treatment Scheme: Stage 1 — *Primary treatment*: Screening removes fibre and lint; equalization tank buffers pH and flow; coagulation-flocculation with alum or ferric chloride removes colour and suspended solids; sedimentation in clarifier. Stage 2 — *Secondary biological treatment*: ASP or UASB + aerobic system reduces BOD/COD. Note: most synthetic dyes are poorly biodegradable, so biological treatment alone is insufficient. Stage 3 — *Tertiary treatment*: Activated carbon adsorption removes residual colour and trace organics; ozonation or UV treatment destroys chromophore groups of dye molecules (decolourisation); Nanofiltration or Reverse Osmosis for desalination and water recovery.

Zero Liquid Discharge (ZLD): ZLD is a water management strategy where no liquid effluent is discharged outside the plant boundary. All treated water is recycled back into the process. Residuals are concentrated to solid/semi-solid form using evaporators and crystallisers. India's National Green Tribunal (NGT) and several State Pollution Control Boards have made ZLD mandatory for textile units in critically polluted areas.

Benefits of ZLD: (a) Zero water pollution of nearby rivers and groundwater. (b) Conservation of freshwater — recovered water reused in dyeing process. (c) Recovery of salt (NaCl) — economic benefit. (d) Compliance with environmental regulations. **Challenges:** High capital and operating cost; high energy consumption of evaporators; management of solid crystallised residue.

Q4. Explain in detail the sources, effects, and control measures for micro-plastic pollution in water bodies.

Introduction: Micro-plastics are plastic particles less than 5 mm in size. They have been detected in oceans, rivers, groundwater, drinking water, food, and even human blood and organs, representing a pervasive new class of environmental pollutants.

Sources: (1) Primary sources — manufactured at micro-scale: microbeads in cosmetics, plastic pellets (nurdles — raw material for plastic manufacturing). (2) Secondary sources — fragmentation of larger plastics by UV radiation, wave action, and mechanical abrasion: plastic bags, bottles, packaging. (3) Synthetic textile fibres released during laundry (up to 700,000 fibres per wash cycle). (4) Tyre wear particles — billions of particles enter stormwater drains. (5) Degrading plastic films used in agriculture.

Pathways into Water: Stormwater runoff → rivers → oceans; WWTP effluent (conventional treatment removes only 70–80% of micro-plastics); atmospheric deposition.

Effects: (a) Marine organisms ingest micro-plastics, causing gut blockage, false satiety, and reduced feeding. (b) Micro-plastics are vectors for POPs (persistent organic pollutants) — "Trojan horse effect." (c) Nano-plastics (<100 nm) penetrate cell membranes, blood-brain barrier, and placenta. (d) Plastic additives (BPA, phthalates) cause endocrine disruption. (e) Antibiotic resistance genes (ARGs) accumulate on micro-plastic surfaces. (f) Humans ingest approximately 5 g plastic/week through food, water, and air.

Control: Ban on microbeads in cosmetics (implemented in USA, UK, Canada, India). Extended Producer Responsibility (EPR) for plastic packaging. Improved wastewater treatment (fine filtration, membrane bioreactors) to capture micro-plastics. Better waste management to prevent plastic entering environment. Consumer awareness and reduction of single-use plastics.

Q5. Describe acid mine drainage — its formation, chemical reactions involved, environmental impacts, and treatment.

Definition: Acid Mine Drainage (AMD) is acidic, metal-rich water formed when iron sulphide minerals (especially pyrite, FeS_2) are exposed to air and water during and after mining operations. It is considered one of the most serious and long-lasting forms of water pollution globally.

Formation Mechanism: Pyrite oxidation proceeds in stages: (1) Chemical oxidation of pyrite by O_2 and water produces ferrous iron (Fe^{2+}) and sulphuric acid. (2) Fe^{2+} is oxidised to Fe^{3+} (ferric iron) — this step greatly accelerated by the bacterium *Acidithiobacillus ferrooxidans*. (3) Fe^{3+} oxidises more pyrite in a self-propagating cycle — the reaction becomes autocatalytic. (4) At low pH, heavy metals (Cu, Zn, Pb, Cd, As, Hg) dissolve from surrounding rock.

Characteristic Features: pH 2–4; bright orange/red precipitate of iron hydroxide ("yellow boy" or "orange slime") coating stream beds; metallic taste; elevated heavy metal concentrations; lifeless water (no fish or invertebrates).

Environmental Impacts: (a) Kills all aquatic life due to extreme acidity and metal toxicity. (b) Contaminates drinking water wells near mines. (c) Destroys agricultural soils if AMD-affected water is used for irrigation. (d) Infrastructure corrosion — pipelines, concrete structures. (e) Persists for centuries after mine closure — legacy pollution.

Treatment: Active — Lime neutralization raises pH to 8–9; metals precipitate as hydroxides and are filtered. Passive — Anoxic limestone drains (ALD); constructed wetlands (plants and sulphate-reducing bacteria neutralise acid and precipitate metals); permeable reactive barriers. Prevention — Mine sealing, backfilling, oxygen-barrier covers over tailings piles.

Q6. Compare aerobic and anaerobic wastewater treatment processes. Explain the stages of anaerobic digestion and its advantages.

Overview: Biological wastewater treatment uses microorganisms to degrade organic matter. Aerobic treatment requires oxygen; anaerobic treatment occurs in its absence.

Stages of Anaerobic Digestion (4 Stages): (1) **Hydrolysis:** Complex organic polymers (carbohydrates, proteins, fats) are broken down by extracellular enzymes into simpler monomers — sugars, amino acids, fatty acids, glycerol. (2) **Acidogenesis:** Monomers are fermented by acidogenic bacteria to produce volatile fatty acids (VFAs — propionate, butyrate, acetate), H_2 , and CO_2 . pH drops; this stage is called the "acid phase." (3) **Acetogenesis:** Acetogens convert VFAs to acetate (CH_3COO^-), H_2 , and CO_2 . Syntrophic relationship with methanogens maintains low H_2 partial pressure. (4) **Methanogenesis:** Methanogens (strict anaerobes) convert acetate and H_2/CO_2 to methane (CH_4) and CO_2 — biogas. This is the slowest and rate-limiting step.

Comparison (see table above for detailed comparison): Aerobic processes are faster, produce more sludge, and require energy for aeration. Anaerobic processes are slower to start, produce valuable biogas, generate less sludge, and are more suitable for high-strength (high-BOD) industrial wastewaters.

Advantages of Anaerobic Treatment: (a) Biogas (60–70% CH_4) is a renewable energy source — can power the treatment plant itself. (b) Very low sludge production — reduces sludge disposal costs. (c) No aeration needed — lower operating energy costs. (d) Small footprint for UASB reactors. (e) Digestate can be used as nutrient-rich bio-fertilizer. **Disadvantages:** Long start-up, sensitive to temperature and toxicants, incomplete pathogen removal, potential H_2S odour.

Q7. Explain the treatment scheme for rice mill (agro-based industry) wastewater. How can a zero-waste approach be implemented?

Introduction: Rice milling, especially parboiling operations, generates large volumes of dark brown, high-BOD wastewater (5,000–12,000 mg/L BOD) that can devastate receiving water bodies if discharged without treatment. In India, rice mills are classified as "Red category" industries by CPCB due to their high pollution potential.

Sources of Wastewater: Soaking water (paddy soaked in hot water for parboiling) — richest in organic pollutants; steaming condensate; washing water; mill floor wash-down.

Treatment Scheme: Step 1 — *Screening:* Removal of rice husk, fibrous material, and large solids. Step 2 — *Primary clarification:* Settling of starch granules, bran, and heavy suspended solids in primary sedimentation tank. Step 3 — *Anaerobic treatment (UASB or covered lagoon):* Core treatment stage for high-BOD wastewater. UASB reactor is widely used — upward flowing wastewater passes through a granular sludge blanket; methane produced is captured for energy use. BOD reduced by 70–80%. Step 4 — *Aerobic polishing (ASP or oxidation pond):* Remaining BOD and ammonia oxidised aerobically. Step 5 — *Constructed wetland or slow sand filter:* Low-cost, eco-friendly final polishing before land disposal or irrigation.

Zero-Waste Concept: (1) Biogas from UASB → used for steam generation in parboiling (energy self-sufficiency). (2) Digestate (anaerobic sludge) → bio-fertilizer for rice farms. (3) Rice husk → combustion fuel for steam boiler, or silica extraction for cement. (4) Rice bran → edible bran oil

extraction (high-value product). (5) Treated wastewater → irrigation of paddy fields (N and P recovery). This circular economy approach dramatically reduces environmental impact while improving the mill's economic viability.

7. Quick Revision — Key Facts & Definitions

Term	Key Point
Eutrophication	Excess nutrients → algal bloom → DO depletion → fish kill
AMD	Pyrite + O_2 + H_2O → sulphuric acid; pH 2–4; orange "yellow boy" precipitate
Micro-plastics	Particles <5 mm; nano-plastics <100 nm; ~5 g/person/week ingested
BOD	Biological Oxygen Demand — O_2 needed to oxidise organic matter in 5 days at 20°C
Primary treatment	Physical — screening, grit removal, sedimentation; removes 30–40% BOD
Secondary treatment	Biological (aerobic/anaerobic); removes 85–95% BOD
Tertiary treatment	Advanced — nutrient removal, disinfection, RO; produces near-potable water
Aerobic digestion	With O_2 ; products: CO_2 + H_2O ; high sludge
Anaerobic digestion	Without O_2 ; 4 stages; products: CH_4 + CO_2 (biogas); low sludge
ZLD	Zero Liquid Discharge — no effluent outside plant; mandatory for textiles in India
UASB	Upflow Anaerobic Sludge Blanket — compact anaerobic reactor; used in India
Parboiling effluent	Extremely high BOD (5,000–12,000 mg/L); treated by anaerobic + aerobic
Chromium (tanning)	Cr(VI) — carcinogenic, mutagenic; precipitated as $Cr(OH)_3$ at pH 8–9
Biomagnification	Concentration of pollutants increases up the food chain (e.g., DDT)

Best of luck with your exams! Master the treatment flow diagrams, remember the BOD removal percentages at each stage, and be ready to compare aerobic vs anaerobic processes — these are examiner favourites. ■